

Changyang Li

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Education

George Mason University

PhD in Computer Science
Advisor: Lap-Fai (Craig) Yu

Aug 2019 - June 2024

University of Virginia

MS in Computer Science

Aug 2017 - May 2019

Beijing Institute of Technology

BS in Computer Science

Aug 2013 - June 2017

Experience

GoerTek Alpha Labs

Research Scientist

Scene perception, generation and reconstruction, human-scene interaction.

Feb 2025 - Current

Meta

Computer Vision Engineer

Parametric human body/hand model, human-object interaction.

June 2024 - Jan 2025

Design Computing and Extended Reality (DCXR) Group

Graduate Research Assistant

Interactive XR system, human-scene interaction, human-centered scene perception.

Aug 2019 - June 2024

Mentor: Lap-Fai (Craig) Yu

Center for Vision, Cognition, Learning, and Autonomy

Research Intern

May 2018 - Aug 2018

Mentor: Song-Chun Zhu

Selected Projects

Scene Perception and Visual Localization

- Developed a scene understanding pipeline using object detection, semantic segmentation, and scene graph modeling techniques for complex indoor environments.
- Presented a visual localization system by fusing semantic features with geometric constraints.

Scene Reconstruction and Content Generation

- Built a 3D reconstruction framework with extended features including 3D consistent instance/semantic segmentation, and 3D/4D Gaussian Splatting representations.
- Utilized generative models (diffusion, flow matching) to synthesize dynamic 3D content, automatically populating reconstructed environments with semantically consistent assets.

Synthetic Human-Object Interaction

- Designed a parametric motion generation system, synthesizing physically plausible human-scene interactions (e.g., sitting, grasping) based on spatial constraints.
- Optimized large-scale human motion datasets to train interaction policies, enabling virtual avatars to adaptively interact with diverse 3D objects.

Context-Aware and Interactive XR System

- Built adaptive XR systems that dynamically accommodates virtual content based on scene semantics and users' inputs and interactions.
- Integrated vision-language models, scene understanding techniques, and scene graph modeling methods, enabling virtual characters to navigate and interact intelligently with physical environments in XR settings.

Selected Publications

Crafting Dynamic Virtual Activities with Advanced Multimodal Models

Changyang Li, Qingan Yan, Minyoung Kim, Zhan Li, Yi Xu, Lap-Fai Yu

2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025

Synthesizing User-Centered Flying Creature Animation Paths for Outdoor Augmented Reality Experiences

Minyoung Kim, Rawan Alghofaili, **Changyang Li**, Lap-Fai Yu

SIGGRAPH 2024 Conference Papers

Generating Activity Snippets by Learning Human-Scene Interactions

Changyang Li, Lap-Fai Yu

ACM Transactions on Graphics (Proceeding of SIGGRAPH 2023)

Location-Aware Adaptation of Augmented Reality Narratives

Wanwan Li*, **Changyang Li***, Minyoung Kim, Haikun Huang, Lap-Fai Yu

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems

Optimizing Product Placement for Virtual Stores

Wei Liang, Luhui Wang, Xinzhe Yu, **Changyang Li**, Rawan Alghofaili, Yining Lang, Lap-Fai Yu

2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

Interactive Augmented Reality Storytelling Guided by Scene Semantics

Changyang Li, Wanwan Li, Haikun Huang, Lap-Fai Yu

ACM Transactions on Graphics (Proceeding of SIGGRAPH 2022)

Synthesizing Scene-Aware Virtual Reality Teleport Graphs

Changyang Li, Haikun Huang, Jyh-Ming Lien, Lap-Fai Yu

ACM Transactions on Graphics (Proceeding of SIGGRAPH Asia 2021)

Learning Virtual Grasp with Failed Demonstrations via Bayesian Inverse Reinforcement Learning

Xu Xie*, **Changyang Li***, Chi Zhang, Yixin Zhu, Song-Chun Zhu

2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2019

Earthquake Safety Training through Virtual Drills

Changyang Li, Wei Liang, Chris Quigley, Yibiao Zhao, Lap-Fai Yu

IEEE Transactions on Visualization and Computer Graphics (Special Issue on IEEE VR 2017)

Patents

Systems and methods for facilitating navigation in space

Changyang Li, Lap Fai Yu

US Patent 12,254,574, 2025

Academic Services and Activities

Program Committee

AAAI, IEEE VR, IEEE ISMAR

Paper Reviewer

- **Conference:** SIGGRAPH, SIGGRAPH Asia, CHI, UIST, Pacific Graphics, IEEE VR, IROS
- **Journal:** ACM Transactions on Graphics, IEEE TVCG, Computer Graphics Forum, IEEE T-RO

Invited Talks

2022 IEEE Metaverse Seminar, 2024 Autodesk 'Research Connections' Talk Series